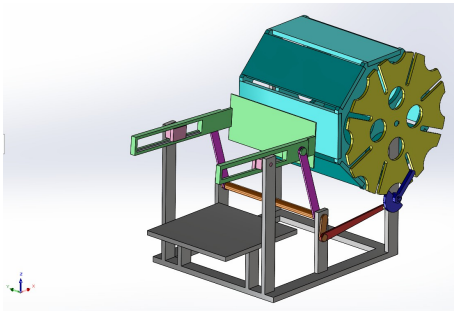
Pick-and-place mechanism

Design and kinematic analysis

SolidWorks / SolidWorks Motion | Academic project



Pick-and-place mechanism assembly modelled in SolidWorks.

1 DOF

motion coordinated by a single drive

OBJECTIVE

Create a virtual model of an object-transfer mechanism and study the trajectories, velocities and accelerations of its main components.

MY CONTRIBUTION

I carried out the CAD modelling, SolidWorks Motion simulation, and generation and interpretation of the kinematic plots. I sized the four-bar linkage and Geneva mechanism jointly with my teammate.

ENGINEERING APPROACH

- Four-bar linkage sizing and assembly modelling in SolidWorks.
- Study of the synchronised motion of the plate and rotating housing, using an eight-slot Geneva mechanism.
- Motion analysis at 15 rpm and generation of graphs and trajectories for the plate, rocker and Geneva wheel.

RESULTS

The model describes a system with one degree of freedom. The analysis documents intermittent 45-degree indexing of the Geneva wheel and 90-degree rotation of the plate, coordinated by the four-bar linkage.

SCOPE AND LIMITATIONS

Kinematic study on a virtual model. The presentation identifies acceleration irregularities caused by contact modelling; this does not constitute physical validation of the mechanism.

PROJECT TEAM

Davide Calvelli / Francesco Paolucci